

OLYMPIC INSIGHTS
USING DATA ANALYTICS

A Power BI Navigational Dashboard Case Study for the International Olympic Committee (IOC)

Tracking Athlete Performance • Medal Trends • Sport-wise Dominance • Forecasting

Presented by

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Data Analytics Project – Power BI | September 2026

BUSINESS CONTEXT

Problem Statement & Objective

The International Olympic Committee (IOC) needs deeper, data-driven visibility into athlete performance, country-wise medal trends, and sport-specific dominance across more than a century of Olympic Games — to guide strategic planning and athlete development.

1

TRACK

Track historical achievements of athletes, countries and sports across every Olympic edition since 1896.

2

EVALUATE

Evaluate performance KPIs — medal efficiency, dominance, participation and host advantage.

3

FORECAST

Forecast future outcomes to guide funding, athlete development and host-bidding decisions.

WHY THIS DASHBOARD

Purpose of the Dashboard

To turn 128 years of scattered Olympic records into one interactive, navigable Power BI tool — so IOC leadership and National Olympic Committees (NOCs) can move from static year-end reports to real-time, self-serve strategic decisions.

1	Which countries and athletes dominate, and by how much?	*Global Medal Overview & Athlete Spotlight*
2	Is a country's medal output efficient relative to its athlete pool?	*Country Insights*
3	Which sports deliver the most medals and deserve investment?	*Sport & Event Analysis*
4	Does hosting the Games actually boost medal performance?	*Host Country Analysis*
5	How is gender balance and athlete age evolving over time?	*Gender & Demographics*

DATA FOUNDATION

Dataset Overview

Six linked tables sourced from historical Olympedia records — covering every Summer & Winter Games from 1896 to 2022 (plus scheduled future editions).

Olympic Athlete Bio

155,861 rows

Athlete identity, sex, birth date, height, country

Olympic Athlete Event Results

316,834 rows

Every athlete's result & medal per event, per edition

Olympic Games Medal Tally

1,807 rows

Country-wise Gold / Silver / Bronze per edition

Olympic Results

7,394 rows

Event-level result detail, dates & venue narrative

Olympics Country

235 rows

NOC code to country name mapping

Olympics Games

64 rows

Edition, host city, host NOC, dates

Scope: 1896 – 2022 | 112 Sports | 964 Distinct Events | 231 Participating Countries (NOCs) | 55 Editions Held

SOLUTION DESIGN

Dashboard Architecture & Navigation

A single Power BI file with 6 linked pages, connected by on-screen navigation buttons — so any user can move between views in one click.

1

PAGE 1

Global Medal Overview

Headline KPIs & top-country medal leaderboard across all Games

2

PAGE 2

Country Insights

Deep-dive into one nation's athletes, efficiency & top sports

3

PAGE 3

Athlete Spotlight

Individual athlete achievements, longevity & demographics

4

PAGE 4

Sport & Event Analysis

Which sports & events drive medal volume and participation

5

PAGE 5

Host Country Analysis

Quantifies the home-ground advantage of hosting the Games

6

PAGE 6

Gender & Demographics

Gender balance and age-profile evolution over 128 years

PAGE 1

Global Medal Overview

Headline KPIs and country leaderboard across all Olympic Games, 1896–2022

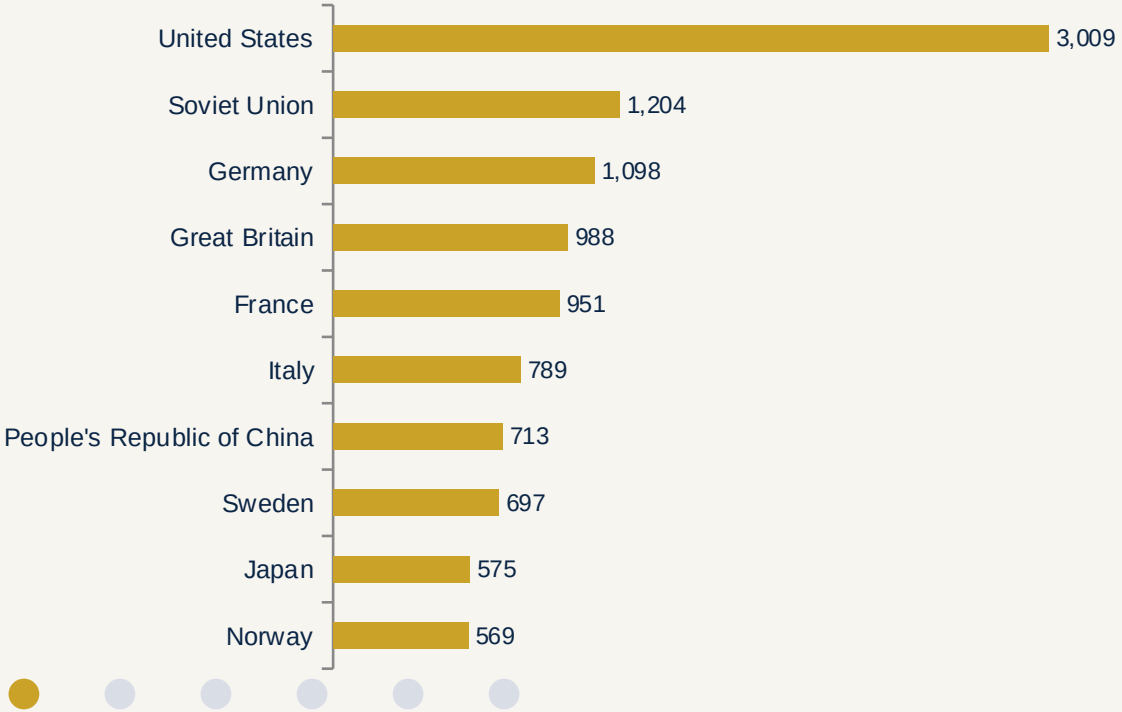
20,654
TOTAL MEDALS AWARDED

231
COUNTRIES PARTICIPATED

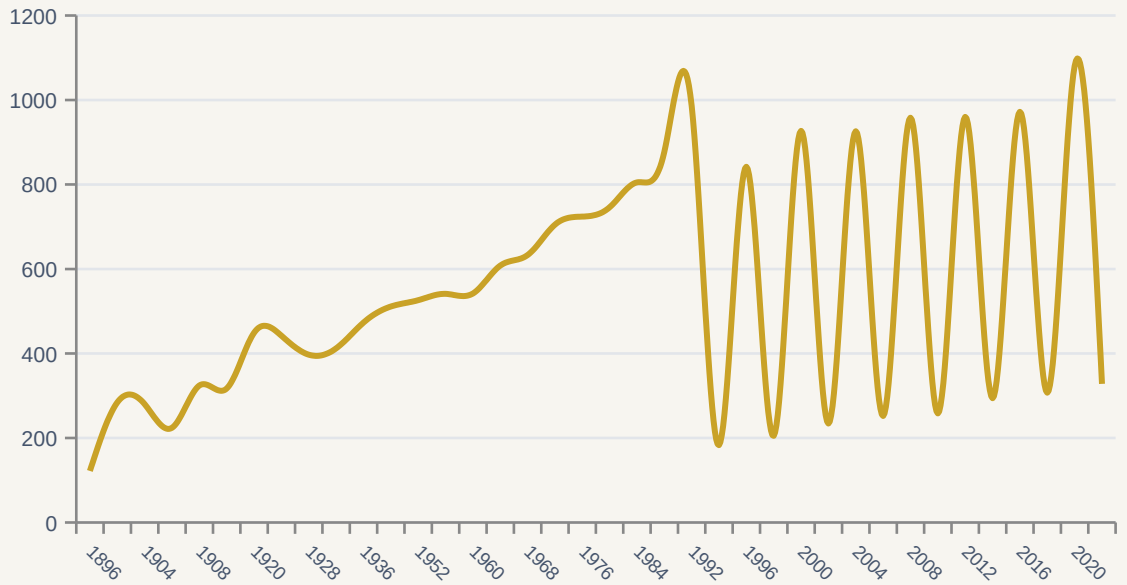
55
OLYMPIC EDITIONS HELD

USA
TOP COUNTRY – 3,009 MEDALS

Top 10 Countries by Total Medals



Total Medals Awarded per Edition (1896–2022)



USA leads with 3,009 career medals — 2.5× the second-ranked nation (USSR, 1,204). Medal volume has grown ~9× since 1896 as more nations & events joined.

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PAGE 2

Country Insights

Deep-dive lens — illustrated here for India; the page filters live to any country in Power BI

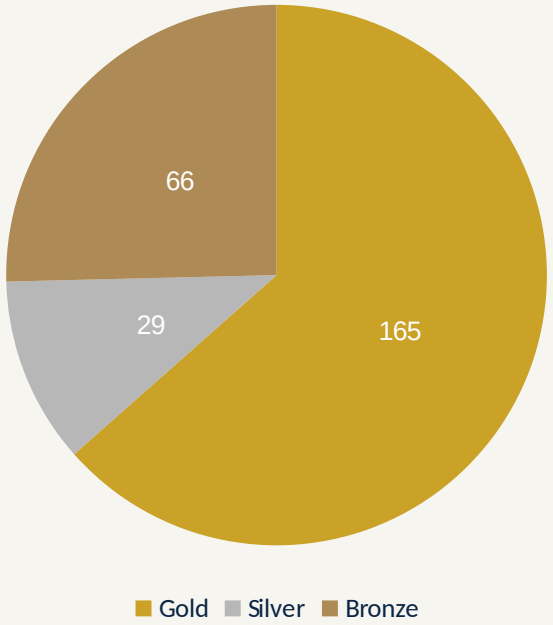
1,055
TOTAL ATHLETES (INDIA)

260
TOTAL MEDALS WON

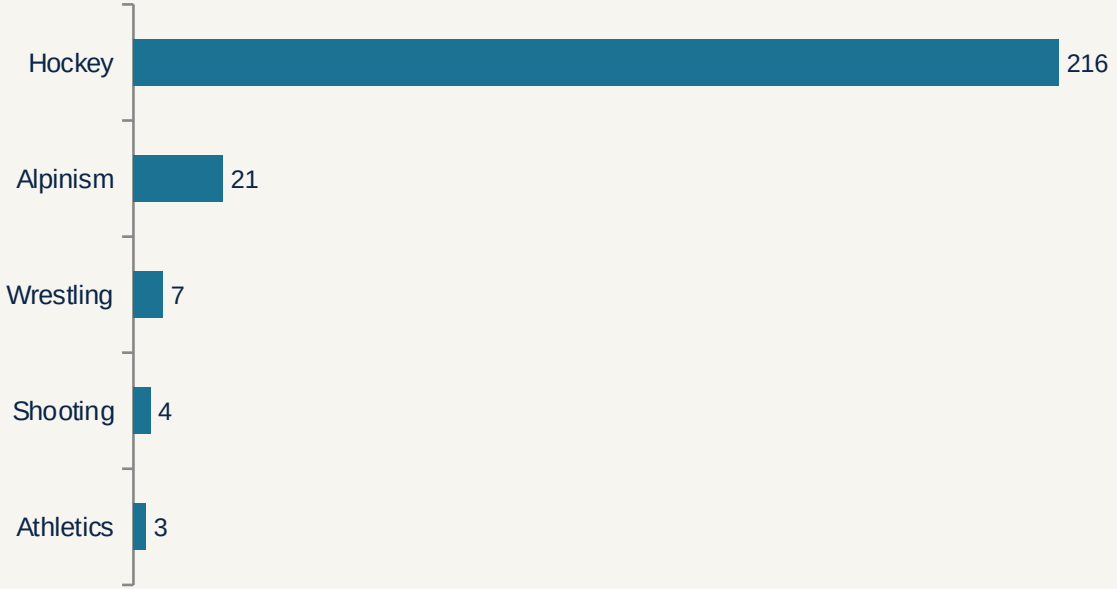
24.64
MEDAL EFFICIENCY /100 ATHLETES

Hockey
BEST SPORT – 216 MEDALS

Medal Distribution (India)



Best Performing Sports (India)



Hockey alone contributes 83% of India's Olympic medals — a sign of strong tradition but heavy concentration risk.

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PAGE 3

Athlete Spotlight

Individual achievement, longevity and demographics across 155,861 Olympians

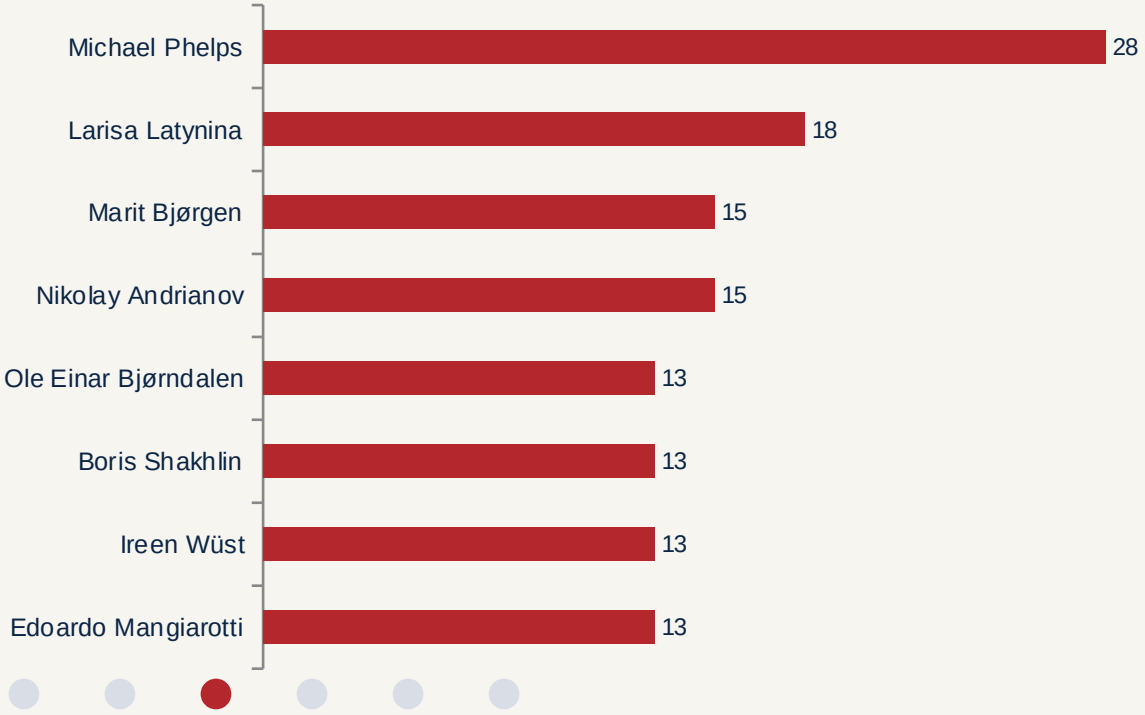
155,867
TOTAL ATHLETES

31,585
MEDAL-WINNING ATHLETES

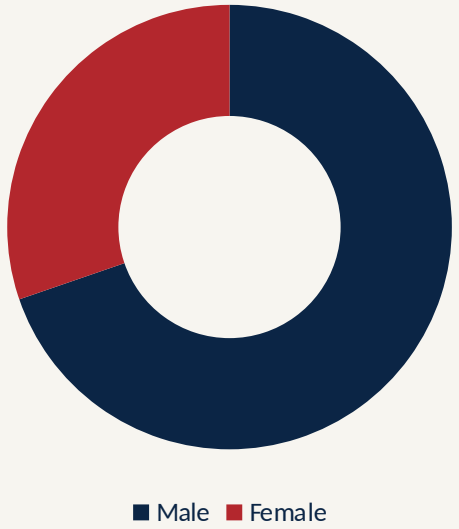
20%
ATHLETES WHO EVER MEDALLED

26.4 yrs
AVG. AGE AT MEDAL WIN

Top Athletes by Total Medals (All-Time)



Gender Split of Medal Winners



Michael Phelps holds the all-time record with 28 medals. Only ~20% of all Olympians ever win a medal — underscoring the value of scouting proven, multi-Games performers.

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PAGE 4

Sport & Event Analysis

Which sports and events drive medal volume and athlete participation

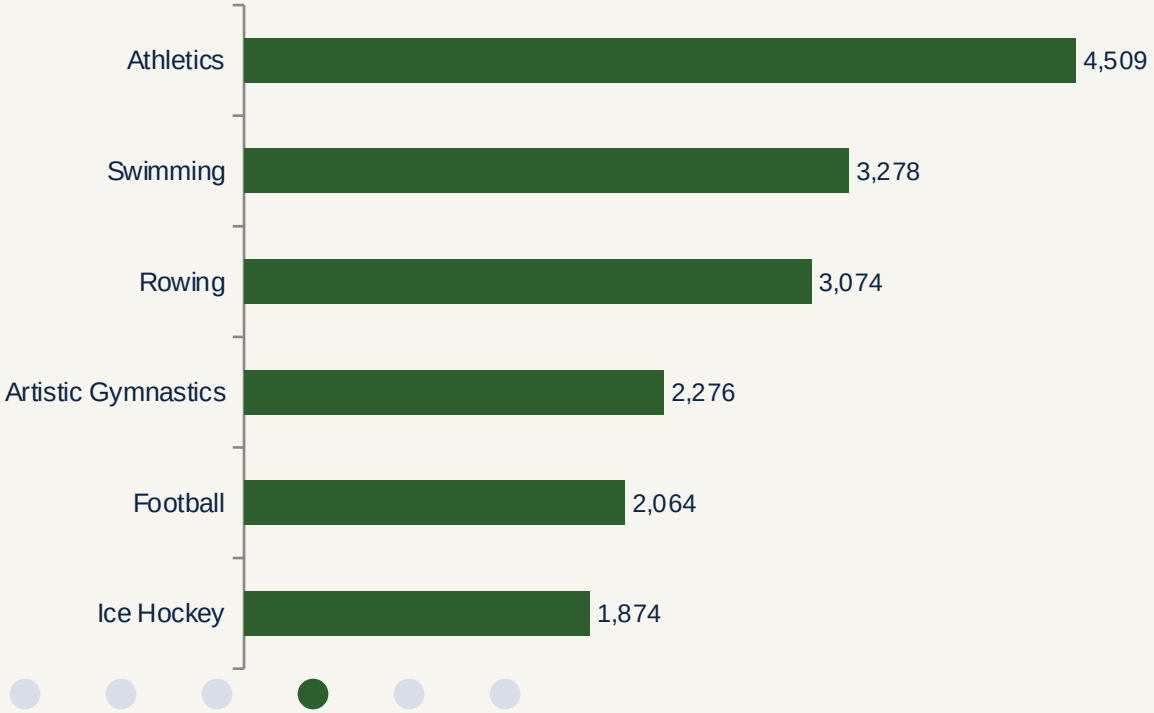
112
TOTAL SPORTS

964
DISTINCT EVENTS

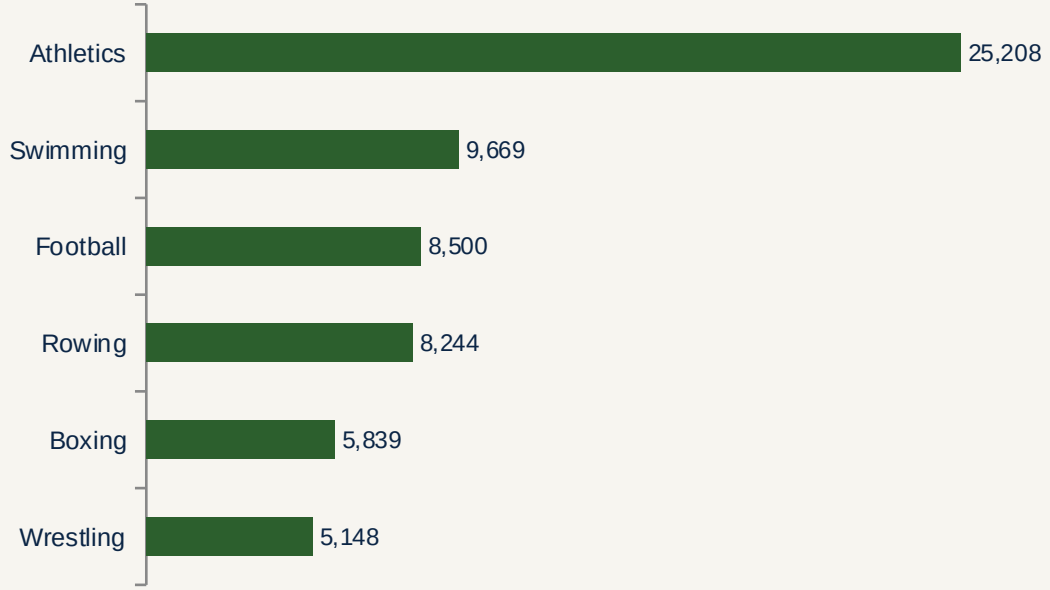
Athletics
TOP SPORT BY MEDALS

4,509
ATHLETICS – MEDALS AWARDED

Top Sports by Medals Awarded



Sport-wise Athlete Participation



Athletics & Swimming together account for ~38% of all medals ever awarded — the two clearest "medal-factory" sports.

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PAGE 5

Host Country Analysis

Quantifying the home-ground advantage of hosting the Olympic Games

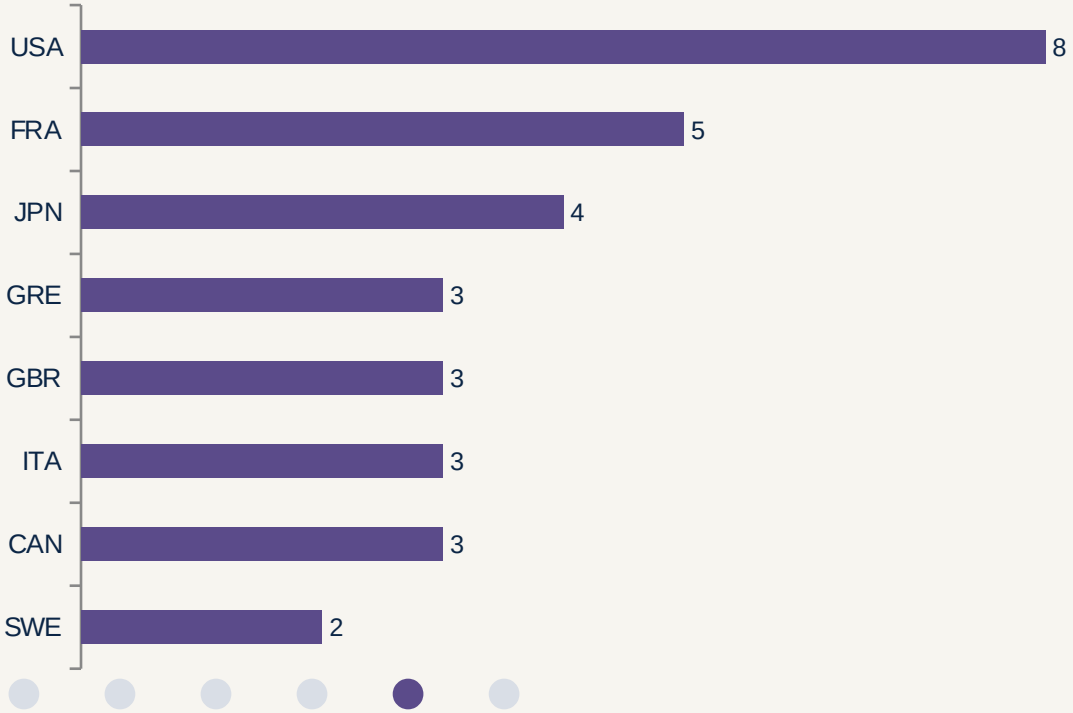
USA (8×)
MOST OLYMPICS HOSTED

42
AVG. MEDALS IN HOST YEAR

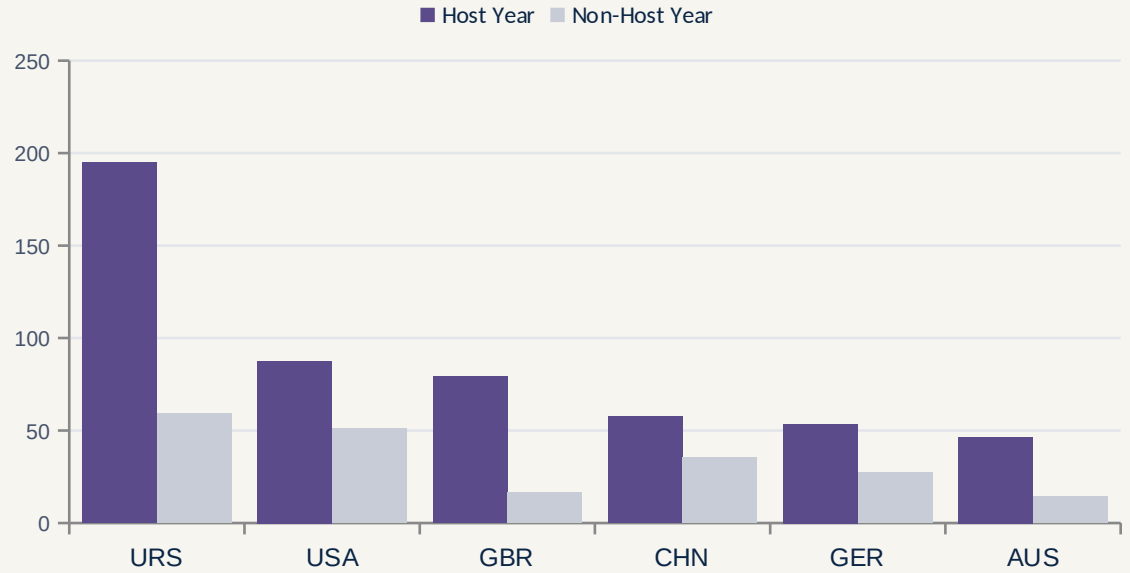
10.5
AVG. MEDALS – NON-HOST YEAR

4.0×
HOST-YEAR MEDAL BOOST

Total Olympics Hosted per Country



Host-Year vs Non-Host-Year Avg. Medals



Across all host nations, average medal haul is 4× higher (42.0) in the year a country hosts vs. other years (10.5) — the "home advantage" is real and measurable.

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PAGE 6

Gender & Demographics

Evolution of gender balance and athlete age profile, 1896–2020

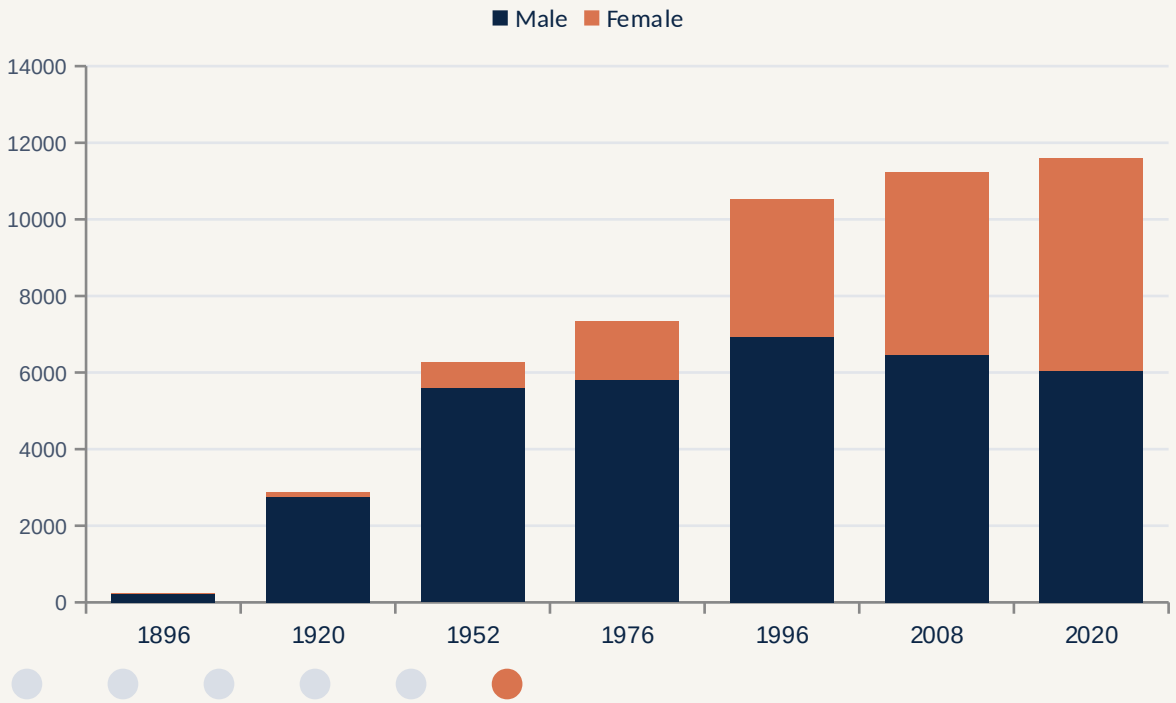
48%
FEMALE SHARE – 2020 GAMES

2 → 5,539
FEMALE ATHLETES (1896→2020)

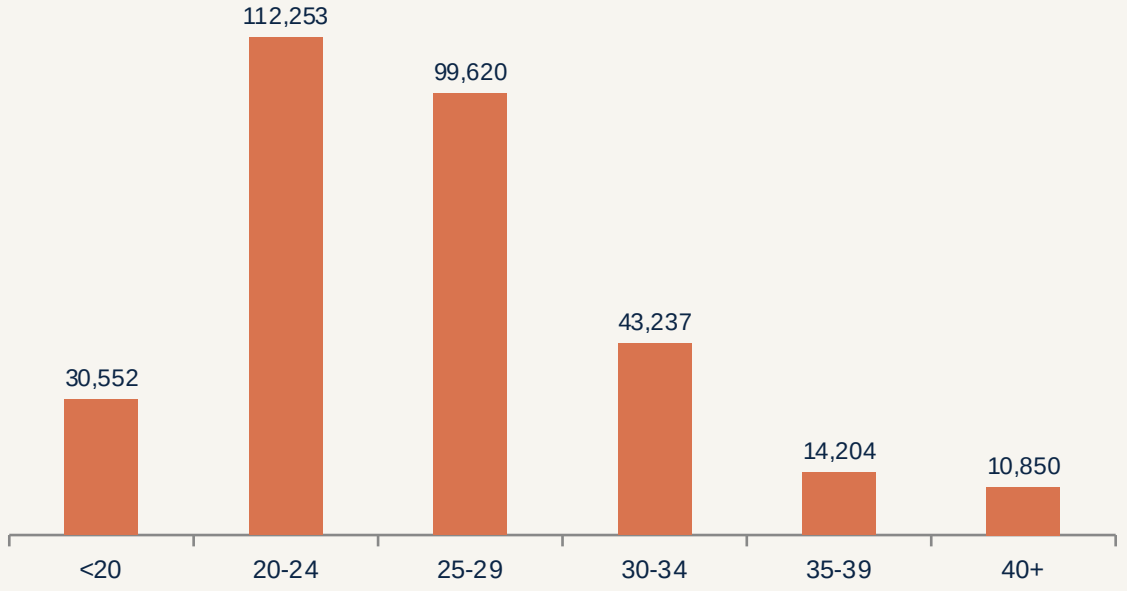
11 yrs
YOUNGEST MEDAL WINNER

69 yrs
OLDEST MEDAL WINNER

Male vs Female Athletes Over the Years



Participation by Age Group (All-Time)



Female participation grew from just 2 athletes (1896) to over 5,500 (2020) — near gender parity. Most medals are won between ages 20–29.

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FROM INSIGHT TO ACTION

Business Decisions This Project Enables

Six pages, six decision levers — turning historical Olympic data into strategic choices for the IOC and NOCs.

PAGE 1 — GLOBAL MEDAL OVERVIEW

Benchmark national competitiveness targets against the true all-time medal leaderboard.

PAGE 2 — COUNTRY INSIGHTS

Diversify sport-wise investment where medal output is concentrated in 1–2 sports (e.g., India & Hockey).

PAGE 3 — ATHLETE SPOTLIGHT

Prioritize scouting & retention of proven multi-Games performers over one-time medalists.

PAGE 4 — SPORT & EVENT ANALYSIS

Direct funding & coaching pipelines toward high medal-density sports like Athletics & Swimming.

PAGE 5 — HOST COUNTRY ANALYSIS

Build the ROI case for Olympic bids using the measured 4× host-year medal advantage.

PAGE 6 — GENDER & DEMOGRAPHICS

Sustain women's-sport investment; near parity reached — focus next on remaining gap sports.

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HOW IT WAS BUILT

Tools & Methodology

1

1. Data Preparation

Power Query cleaned & merged 6 CSV tables (~480,000 rows) — fixed NOC code changes (URS/RUS, FRG/GER), parsed birth dates, handled nulls in age/medal fields.

2

2. Data Modeling

Built a star schema: Games & Country as dimension tables; Athlete Event Results & Medal Tally as fact tables, linked on edition_id, athlete_id & country_noc.

3

3. DAX Measures

Authored custom measures for Medal Efficiency, Host-Year Flag, YoY Medal Growth, Age Buckets and Gender Share — see next slide.

4

4. Dashboard Build

Designed 6 navigational pages in Power BI Desktop with bookmark-based buttons, slicers (Year, Country, Sport, Gender) and a consistent Olympic colour theme.

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UNDER THE HOOD

Sample DAX Measures Used in the Model

Core DAX formulas powering the KPIs seen on each dashboard page.

Total Medals Awarded

```
Total Medals =
SUM ( Olympic_Games_Medal_Tally[total] )
```

Medal Efficiency (per 100 athletes)

```
Medal Efficiency =
DIVIDE (
    DISTINCTCOUNT
( Event_Results[athlete_id] ),
    [Total Athletes]
) * 100
```

Host-Year Flag

```
Is Host Year =
IF (
    SELECTEDVALUE ( Medal_Tally[country_noc] )
=
        SELECTEDVALUE ( Games[country_noc] ),
    "Host", "Non-Host"
)
```

Host vs Non-Host Avg. Medals

```
Avg Medals (Host vs Non-Host) =
AVERAGEX (
    FILTER ( Medal_Tally, [Is Host Year] =
"Host" ),
    Medal_Tally[total]
)
```

Athlete Age at Medal Win

```
Age At Medal =
DATEDIFF (
    Athlete_Bio[born], Games[start_date], YEAR
)
```

YoY Medal Growth %

```
Medal YoY Growth % =
VAR CurrYear = [Total Medals]
VAR PrevYear =
    CALCULATE ( [Total Medals], PREVIOUS
( Games[year] ) )
RETURN DIVIDE ( CurrYear - PrevYear, PrevYear )
```


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REFLECTION

Challenges & Learnings

Historical NOC Code Changes

Countries like USSR/Russia and West/East Germany changed codes across editions — required careful NOC mapping before aggregating medals by “country”.

Missing Birth Dates

Older athlete records (pre-1920s) often lacked complete birth dates, so age-based analyses were filtered to valid, plausible age ranges (10–70 yrs).

Team vs Individual Medals

The isTeamSport flag had to be respected so team-sport medals weren't over- or under-counted per athlete vs. per country.

Data Volume & Relationships

Modeling relationships across ~480,000 rows across 6 tables required a disciplined star-schema to keep the dashboard fast and accurate.

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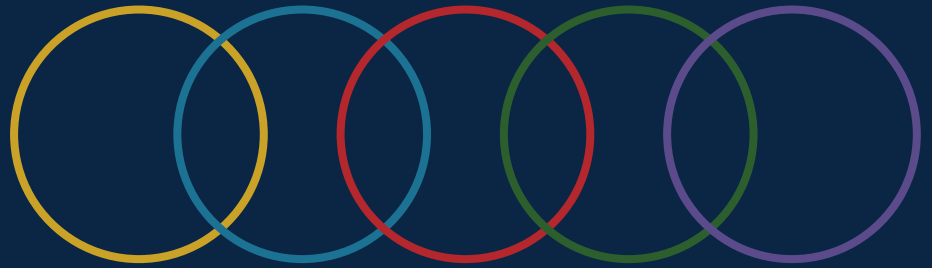
WRAPPING UP

Conclusion & Recommendations

This 6-page navigational Power BI dashboard converts 128 years of Olympic history into a single, decision-ready tool for the IOC — covering medal leadership, country efficiency, athlete performance, sport investment, host advantage and demographic shifts.

- ✓ Adopt this dashboard as a standing tool for annual strategic reviews, refreshed after every Games.
- ✓ Use the Host Country page's 4× medal-boost finding as quantified evidence in future host-bid proposals.
- ✓ Direct athlete-development budgets toward high medal-density sports while diversifying single-sport-dependent nations.
- ✓ Track the Gender & Demographics page each cycle to sustain progress toward full participation parity.

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THANK YOU

Olympic Insights Using Data Analytics — Power BI Case Study

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Vaishnavi Lahoti

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